

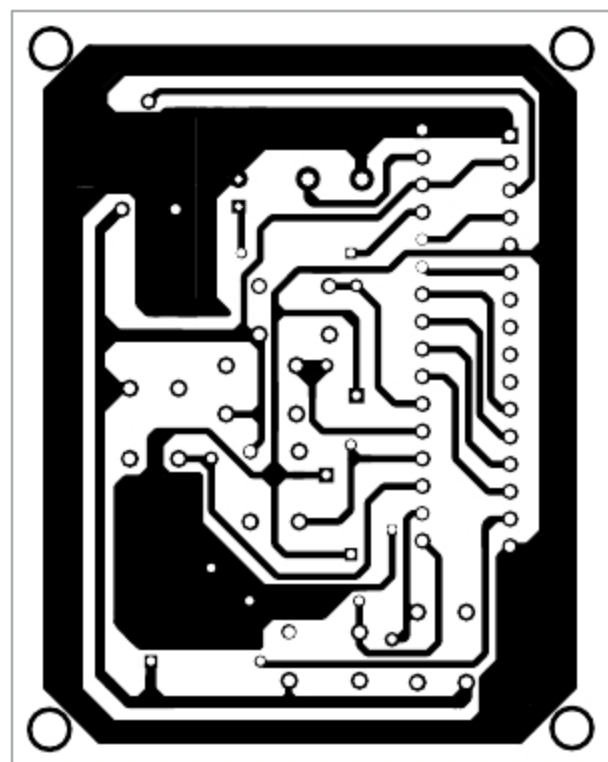


Fig. 3: Actual-size PCB layout of the fastest finger first system

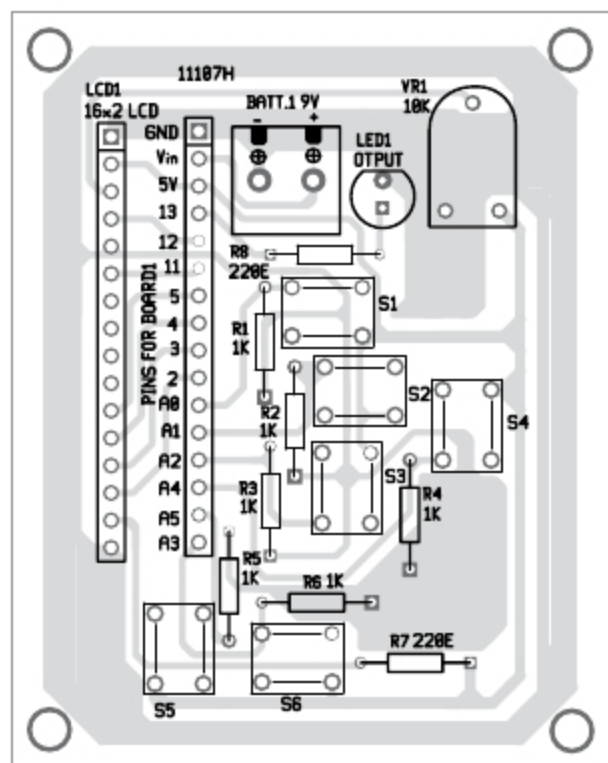


Fig. 4: Components layout for the PCB

pressed key is displayed on the LCD.

Preset VR1 is used to control contrast of the LCD.

LED. Whenever a key is pressed, an LED (LED1), which is connected to digital pin 13 of Arduino, glows for 0.5 seconds.

As observed from Fig. 2, digital pins 12, 11, 5, 4, 3 and 2 of Arduino are connected to RS, E, 4, 5, 6 and 7 pins, respectively, of the LCD. When no keys are pressed, Arduino reads 5V from each column. If more than

PARTS LIST

Semiconductors:

Board1 - Arduino Uno board
LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R6 - 1-kilo-ohm
R7-R8 - 220-ohm
VR1 - 10-kilo-ohm preset

Miscellaneous:

BATT.1 - 9V battery
S1-S6 - Tactile switch
LCD1 - 16x2 alphanumeric LCD

one key is pressed, Arduino reads 0V from the corresponding column of the fastest pressed key while rejecting the readings from other keys. For example, if 1 is pressed before 6, then A0 pin will be read by Arduino.

Software

The source code is written in Arduino programming language. ATmega328/328P is programmed using Arduino IDE software. Select the correct board from Tools→Board menu in Arduino IDE and burn the program (sketch) through the standard USB port in your computer.

Source code (fast_press_fast_print.ino) uses LiquidCrystal.h header file provided by Arduino library for working with the LCD.

lcd.begin(16, 2) function helps configure the LCD.

Serial.begin(9600) function initialises the serial port with a baud rate of 9600.

Construction and testing

An actual-size PCB layout of the fastest finger first system is shown in Fig. 3 and its components layout in Fig. 4. Assemble the components as per the circuit diagram on the PCB and connect the 9V battery. After this, the LCD will display “Start the game” message, and users have to press the switch. Then, it will display the number assigned to the candidate who pressed the switch first. **EFY**



Souvik Kumar Das is passionate about electronics and MCU-based system design



Shibendu Mahata is M.Tech (gold medalist) in instrumentation and electronics engineering from Jadavpur University. He has keen interest in MCU-based real-time embedded signal processing and process control systems

EFY Note

The source code of this project is available for free download at source.efymag.com

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